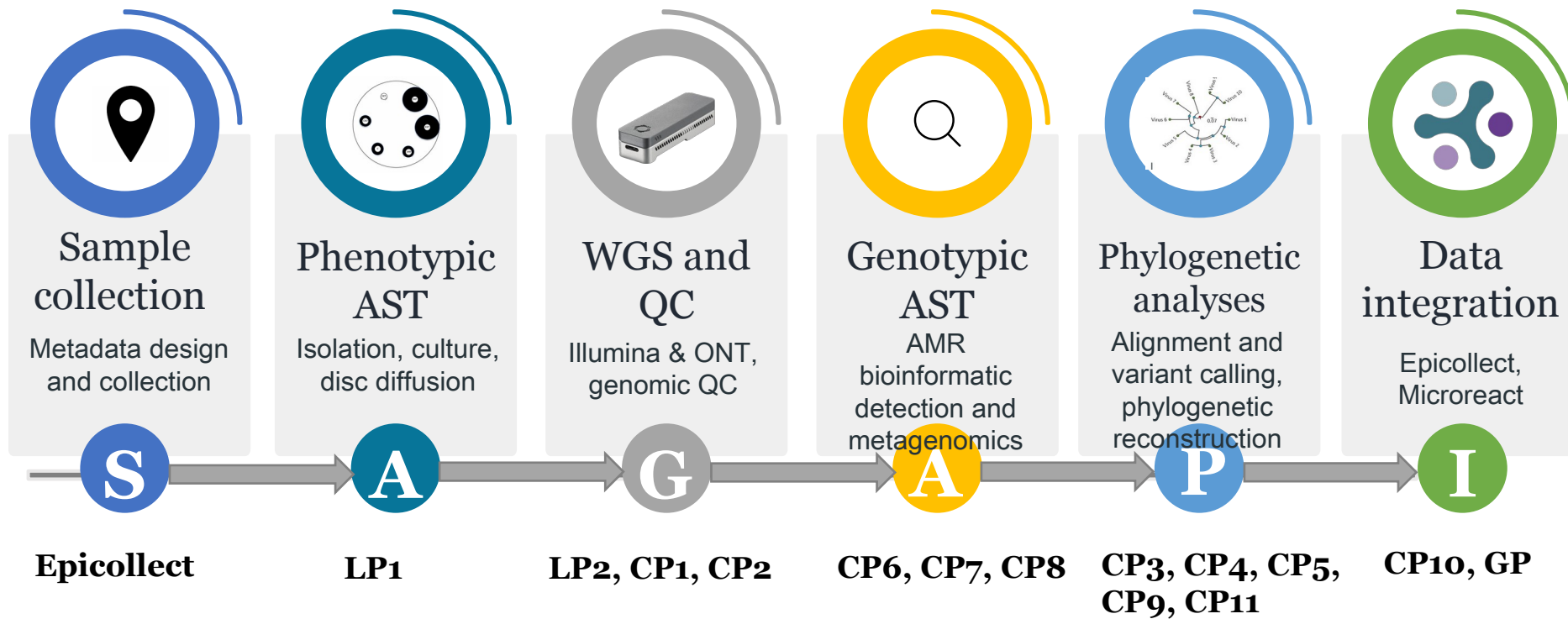


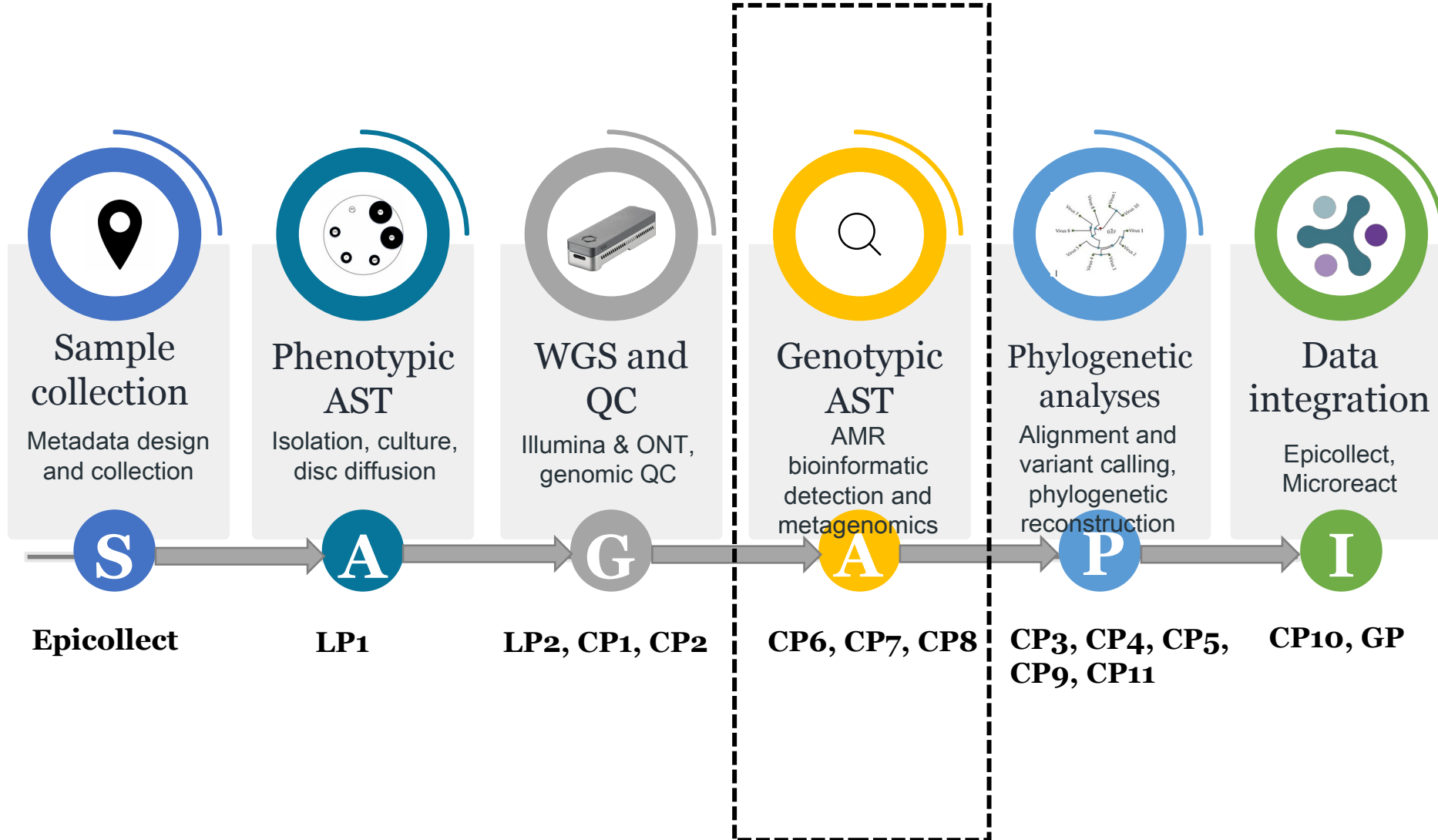
Computational Practical 6

Detecting antimicrobial resistance from bacterial genomes using command-line tools

Antimicrobial Resistance of Bacterial Pathogen Course
Johannesburg, South Africa

10/03/2026





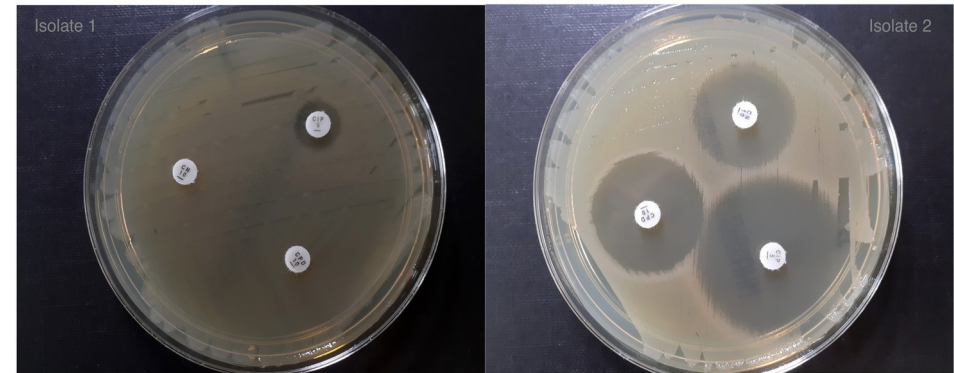
Commercial phenotypic AMR detection assays

Routine culture-based AST

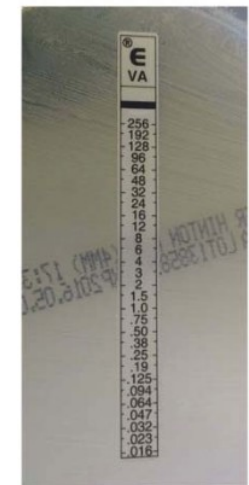
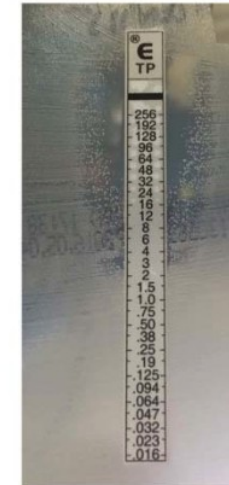
AST is routinely performed using culture-based methods, such as disk diffusion, broth microdilution or gradient diffusion (the E-test).



Broth microdilution



Disk diffusion



E-test

Approaches for AMR genotypic determination

→ Look-up table and rule-based approaches



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<https://doi.org/10.1128/JCM.03117-13>

Prediction of *Staphylococcus aureus* Antimicrobial Resistance by Whole-Genome Sequencing

N. C. Gordon^a, J. R. Price^b, K. Cole^b, R. Everitt^c, M. Morgan^d, J. Finney^a, A. M. Kearns^e, B. Pichon^e, B. Young^a, D. J. Wilson^a, M. J. Llewelyn^b, J. Paul^f, T. E. A. Peto^a, D. W. Crook^a, A. S. Walker^a, T. Golubchik^a

TABLE 2 Panel of housekeeping genes with amino acid variants known to be associated with antimicrobial resistance^a

Antimicrobial agent	Gene	Amino acid substitutions ^b	Reference gene accession no. (nucleotide positions)
Ciprofloxacin	<i>gyrA</i>	S84L, E88K, G106D, S85P, E88G, E88L	BX571857.1 (7005–9668)
	<i>griA</i>	S80F, S80Y, E84K, E84G, E84V, D432G, Y83N, A116E, I45M, A48T, D79V, V41G, S108N	BX571857.1 (1386869–1389271)
	<i>griB</i>	R470D*, E422D*, P451S*, P585S*, D443E*, R444S*	BX571857.1 (1384872–1386869)
Fusidic acid	<i>fusA</i>	A160V*, A376V, A655E, A655P*, A655V*, A67T*, A70V*, A71V*, B434N, C473S*, D189G*, D189V*, D373N*, D463G*, E233Q*, E444K, E444V*, E449K*, F441Y, F652S*, G451V, G452C, G452S, G556S, G617D, G664S, H438N, H457Q, H457Y, L430S*, L456F, L461K, L461S, M161I*, M453I, M651I, P114H, P404L, P404Q, P406L, P478S, Q115L, R464C, R464H, R464S, R659C, R659H, R659L, R659S, R76C*, S416F*, T385N, T387I*, T436I, T656K, V607I, V90A, V90I, Y654N*	BX571857.1 (577685–579766)
Rifampin	<i>rpoB</i>	A473T*, A477D, A477T*, A477V, D471G*, D471Y, D550G, H481D, H481N, H481Y, I527F, I527L*, I527 M*, ins 475H, ins G475*, L466S*, M470T*, N474K*, Q456K, Q468K, Q468L, Q468R, Q565R*, R484H, S463P, S464P, S486L, S529L*	BX571857 (568813–572364)
Trimethoprim	<i>dhfrB</i>	F99Y, F99S, F99I, H31N, L41F, H150R, L21V*, N60I*	BX571857.1 (1464014–1464493)

^a All housekeeping gene sequences were obtained from the genome of reference strain MSSA 476 (7).

^b An asterisk indicates that the amino acid substitution was reported in association with other variants. For the expected effect of each variant/combination on MIC, please see the supplemental data. ins, insertion.

TABLE 1 Panel of genes associated with mobile genetic elements used for BLAST query^a

Antimicrobial agent(s)	Gene ^b	Product ^c	Reference gene accession no. (nucleotide positions)
Penicillin	<i>blaZ</i>	Class A beta-lactamase	BX571856.1 (1913827–1914672)
Methicillin	<i>mecA</i>	Low-affinity PBP2	BX571856.1 (44919–46925)
Erythromycin	<i>msrA</i> *	Erythromycin resistance protein	CP003194 (54168–55634)
Erythromycin and clindamycin	<i>ermA</i>	rRNA adenine N-6-methyltransferase	BA000018.3 (56002–56733)
	<i>ermB</i>	rRNA adenine N-6-methyltransferase	AB699882.1 (4971–5708)
	<i>ermC</i>	rRNA adenine N-6-methyltransferase	HE579068 (7858–8592)
	<i>ermT</i>	23S rRNA methylase	HF583292 (11344–12078)
Tetracycline	<i>tetK</i>	MFS tetracycline efflux pump	FN433596 (69118–70497)
	<i>tetL</i>	MFS tetracycline efflux pump	HF583292 (7713–9089)
	<i>tetM</i>	Ribosomal protection protein	CP002643 (427033–428952)
Vancomycin	<i>vanA</i>	Low-affinity peptidoglycan precursor	AE017171.1
Fusidic acid	<i>fusB</i>	Fusidic acid detoxification	CP003193.1 (1336–1977)
	<i>far</i> *	Ribosome protection protein	AY373761.1 (19072–19713)
Trimethoprim	<i>dhfrA</i>	Insensitive dihydrofolate reductase	CP002120 (2093303–2093788)
	<i>dhfrG</i>	Insensitive dihydrofolate reductase	FN433596 (502263–502760)
Gentamicin	<i>aacA-aphD</i>	6'-aminoglycoside N-acetyltransferase/ 2"-aminoglycoside phosphotransferase	FN433596.1 (2209531–2210970)
Mupirocin (high-level resistance)	<i>mupA</i>	Isoleucyl-tRNA synthetase	HE579068 (2157–5231)
	<i>mupB</i>	Isoleucyl-tRNA synthetase	JQ231224

^a Presence of the gene correlates with phenotypic resistance. Reference sequences were obtained from published sequences from human clinical isolates.

^b An asterisk indicates that the gene was added in v2.0.

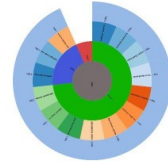
^c MFS, major facilitator superfamily.

Gordon, N. C. *et al.* Prediction of *Staphylococcus aureus* Antimicrobial Resistance by Whole-Genome Sequencing. *Journal of Clinical Microbiology* 52, 1182–1191 (2014).

Available databases and tools

arpcard/**rgi**

Resistance Gene Identifier (RGI). Software to predict resistomes from protein or nucleotide data, including metagenomics data, based on homology and...



👤 13 Contributors
🕒 25 Issues
★ 240 Stars
🍴 61 Forks



<https://github.com/arpcard/rgi>

ncbi/**amr**

amr_finder



👤 5 Contributors
🕒 5 Issues
★ 151 Stars
🍴 26 Forks



<https://github.com/ncbi/amr> (AMRFinderPlus)

cadms/**resfinder**



👤 2 Contributors
🕒 3 Issues
★ 3 Stars
🍴 2 Forks



<https://github.com/cadms/resfinder>

Outbreak strains genomes to analyse

Bacterial strains to be analysed in this practical

Table 1 contains the list of strains to be analysed in this practical from the CPE outbreak we will be investigating this year. Table 2 contains additional strains to be analysed (optionally, if time allows) sourced from key studies on the genomic epidemiology of methicillin-resistant *Staphylococcus aureus* (MRSA) ([Holden et al. 2013](#)) and extensively drug-resistant (XDR) *Salmonella typhi* ([Klemm et al. 2018](#)). In this practical we will use three different command-based tools (AMRFinder, ResFinder and CARD RGI) to identify AMR genetic determinants from whole-genome sequences. In later computational practicals we will explore the epidemiology of outbreaks and epidemic clones of these bacteria, and study the strains investigated in this practical in a broader context. We will also identify the genomic context of AMR genes, for example, if they are carried on mobile genetic elements.

Table 1 CPE strains to be analysed in this practical

Species	Study and origin	Strain Id	Illumina accession	Assembly file name
<i>K. pneumoniae</i>	Roberts et al. 2024, CPE strain	cpe004	ERR4095909	cpe004_Kpn-ST78-NDM1.fasta
<i>E. coli</i>	Roberts et al. 2024, CPE strain	cpe069	ERR5386320	cpe069_Eco-NDM1.fasta

Outbreak strains genomes to analyse

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Table 2 Additional strains to be analysed (optional).

Species	Study and origin	Strain Id	Genome accession	Assembly file name
<i>S. aureus</i>	Holden et al. 2013, Berlin (Germany), 2007, ST22 EMRSA-15	07-02477	ERR017261	ERR017261.assembly.fa
<i>S. aureus</i>	Holden et al. 2013, UK, 2005, ST22 EMRSA-15	HO50960412	HE681097 (GenBank)	HO50960412.fa
<i>S. typhi</i>	Klemm et al. 2018 (ACT), Pakistan, 2016, 4.3.1 (H58) XDR	BL0006	ERR2093245	ERR2093245.assembly.fa
<i>S. typhi</i>	Klemm et al. 2018, Pakistan (2016) – 4.3.1 (H58) pre-XDR	Pak60168	ERR2093329	ERR2093329.assembly.fa

https://github.com/WCSCourses/AMR_2026/tree/main/course_modules_2026/Detecting_resistance/

WCSCourses / AMR_2026

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course_data_2026

course_modules_2025

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Data_access_QC

Detecting_resistance

Anthony_Enterobase BIGSdb N...

Enterobase manual.pdf

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francescoll Add files via upload a8e37ae · 3 minutes ago History

This branch is 130 commits ahead of WCSCourses/AMR_Bacterial_Pathogens:main. Contribute Sync fork

Name	Last commit message	Last commit date
..		
Anthony_Enterobase BIGSdb NCBIpathogendetection.pdf	Add files via upload	2 weeks ago
Enterobase manual.pdf	Add files via upload	2 weeks ago
README.md	Create README.md	2 weeks ago
Summary of genotypic AMR results - CPE strains.docx	Add files via upload	3 minutes ago
computational_practical_6_manual.md	Update computational_practical_6_manual.md	4 hours ago

README.md

When interpreting the results: link antibiotic to antibiotic class

V · T · E	Antibacterials that inhibit nucleic acid (J01E, J01M) [hide]		
Antifolates (inhibit bacterial purine metabolism, thereby inhibiting DNA and RNA synthesis)	DHFR inhibitor	2,4-Diaminopyrimidine (Brodinoprim · Iclaprim [†] · Ormetoprim · Pyrimethamine [#] · Tetroxoprim · Trimethoprim [#])	
	Sulfonamides (DHPS inhibitor)	Short-acting	Sulfaisodimidine · Sulfamethizole · Sulfadimidine · Sulfapyridine (Sulfasalazine) · Sulfafurazole (Acetyl sulfisoxazole) · Sulfanilamide (Prontosil) · Sulfathiazole (Phthalylsulfathiazole, Succinylsulfathiazole) · Sulfathiourea
		Intermediate-acting	Sulfamethoxazole · Sulfadiazine [#] · Sulfamoxole
		Long-acting	Sulfadimethoxine · Sulfadoxine · Sulfalene · Sulfametomidine · Sulfametoxydiazine · Sulfamethoxypyridazine · Sulfaperin · Sulfamerazine · Sulfaphenazole · Sulfamazone
		Other/ungrouped	Maferide · Sulfacetamide · Sulfaclozine · Sulfadicramide · Sulfaguanidine · Sulfametrole · Sulfanitran
	Combinations	Trimethoprim/sulfamethoxazole [#] · Ormetoprim/sulfadimethoxine · Pyrimethamine/dapsone · Pyrimethamine/sulfadoxine	
Quinolones (inhibit bacterial topoisomerase and/or DNA gyrase, thereby inhibiting DNA replication)	Other DHPS inhibitors	Acediasulfone · Dapsone · Solasulfone · Sulfoxone	
	1st generation	Cinoxacin [‡] · Flumequine [‡] · Nalidixic acid [‡] · Oxolinic acid [‡] · Pipemidic acid [‡] · Piromidic acid [‡] · Rosoxacin [‡]	
	Fluoroquinolones	2nd generation	Ciprofloxacin [#] · Ofloxacin · Enoxacin [‡] · Fleroxacin [‡] · Lomefloxacin [‡] · Nadifloxacin [‡] · Norfloxacin [‡] · Pefloxacin [‡] · Rufloxacin [‡]
		3rd generation	Levofloxacin [#] · Balofloxacin [‡] · Grepafloxacin [‡] · Pazufloxacin [‡] · Sparfloxacin [‡] · Temafloxacin [‡] · Tosufloxacin [‡]
		4th generation	Besifloxacin · Delafloxacin · Gatifloxacin · Finafloxacin · Gemifloxacin · Moxifloxacin [#] · Clinafloxacin [‡] · Garenoxacin [‡] · Prulifloxacin [‡] · Sitafoxacin [‡] · Trovafloxacin [‡] /Alatrofloxacin [‡]
		Veterinary	Danofloxacin · Difloxacin · Enrofloxacin · Ibafoxacin · Marbofloxacin · Orbifloxacin · Pradofloxacin · Sarafloxacin [‡]
Anaerobic DNA inhibitors	Newer non-fluorinated	Nemonoxacin	
	Related (DG)	Aminocoumarins: Novobiocin	
	Nitroimidazole derivatives	Metronidazole [#] · Ornidazole · Secnidazole · Tinidazole	
RNA synthesis	Nitrofuran derivatives	Furazolidone [‡] · Nifuroxazide · Nifurtoinol · Nifurzide · Nitrofurantoin [#] · Nitrofurazone	
	Rifamycins/ RNA polymerase	Rifampicin [#] · Rifabutin [#] · Rifapentine [#] · Rifaximin · Rifalazil ^{\$}	
	Liparmycins	Fidaxomicin	

<https://en.wikipedia.org/wiki/Ciprofloxacin>